Project Title

Phishing URL Detection Using Ridge- and Lasso-Regularized Logistic Regression

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ABSTRACT

Phishing is a common cybersecurity threat where attackers create fake websites or misleading URLs to trick users into sharing sensitive information such as passwords, banking details, and personal data. Since phishing URLs can sometimes look similar to genuine URLs, identifying them manually can be difficult. This project, “Phishing URL Detection Using Ridge- and Lasso-Regularized Logistic Regression,” aims to develop a machine-learning-based system that can classify URLs as either phishing or legitimate. The system learns patterns from a dataset containing both phishing and legitimate URLs.

The system extracts important features from URLs, such as URL length, number of special characters, HTTPS usage, presence of an IP address, and suspicious words. These features are used by Logistic Regression to classify the URLs. To reduce overfitting and improve prediction, Ridge and Lasso regularization are applied. Ridge helps control the importance of features, while Lasso can remove less useful features from the model.

The project compares Logistic Regression, Ridge-Regularized Logistic Regression, and Lasso-Regularized Logistic Regression to determine which method performs better. The models are evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The best-performing model can then be used to predict whether a new URL is potentially phishing or legitimate.

Overall, the project demonstrates how machine learning can be used to detect phishing URLs and help users identify potentially unsafe websites. It also provides a practical understanding of how Ridge and Lasso regularization can improve the performance of a classification model.

Signature of the Guide